

Definition. If A is an $n \times n$ matrix in $\mathbb{R}^n$ such that

$$Av = \lambda v, \quad v \neq 0, \quad \lambda \in \mathbb{R},$$

then λ is an **eigenvalue** for A , and v is an **eigenvector** for A .

Question. Is λ real?

$$\lambda = 3, \quad Av = \begin{bmatrix} 9 \\ 18 \end{bmatrix} = \lambda \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Theorem. If $Av = \lambda v$, then $(\lambda^k v)$ solves

$$X_{k+1} = AX_k.$$

Proof.

$$X_{k+1} = \lambda^{k+1}v = A\lambda^k v = \lambda^k Av = \lambda^k \lambda v.$$

Theorem. If $\{v_1, \dots, v_n\}$ is a basis in $\mathbb{R}^n$ and

$$Av_i = \lambda_i v_i, \quad i = 1, 2, \dots, n,$$

then

$$\{(\lambda_1^k v_1), \dots, (\lambda_n^k v_n)\}$$

is a basis in the solution space of

$$X_{k+1} = AX_k.$$

Proof. The solution space of $X_{k+1} = AX_k$ has dimension n , and each $(\lambda_i^k v_i)$ is a solution. To check independence, suppose

$$c_1(\lambda_1^k v_1) + \dots + c_n(\lambda_n^k v_n) = 0.$$

Set $k = 0$. Then

$$c_1 v_1 + \dots + c_n v_n = 0,$$

so $c_1 = \dots = c_n = 0$.

Representing a linear map by a matrix

Let $T : V \rightarrow W$ be a linear map where

$$\dim V = n, \quad \dim W = m.$$

Let

$$\beta = \{v_1, \dots, v_n\}$$

be a basis of V , and let

$$\gamma = \{w_1, \dots, w_m\}$$

be a basis of W . Can we represent T by a matrix?

Define coordinate maps

$$S : V \rightarrow \mathbb{R}^n, \quad S(v) = [v]_\beta,$$

and

$$R : W \rightarrow \mathbb{R}^m, \quad R(w) = [w]_\gamma.$$

The coordinate maps and T fit into the diagram

$$\begin{array}{ccc} V & \xrightarrow{S} & \mathbb{R}^n \\ T \downarrow & & \\ W & \xrightarrow{R} & \mathbb{R}^m. \end{array}$$

Thus

$$S(c_1v_1 + \dots + c_nv_n) = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix},$$

and

$$R(d_1w_1 + \dots + d_mw_m) = \begin{bmatrix} d_1 \\ \vdots \\ d_m \end{bmatrix}.$$

The map

$$RTS^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is linear, so for some matrix A ,

$$RTS^{-1}(x) = Ax.$$

The vector Ae_i is the i th column of A . Since

$$S^{-1}(e_i) = v_i,$$

we have

$$Ae_i = [T(v_i)]_\gamma.$$

Theorem.

$$A = \begin{bmatrix} [T(v_1)]_\gamma & \cdots & [T(v_n)]_\gamma \end{bmatrix}.$$

Example. Let

$$T : \mathbb{R}_{\leq 2}[x] \rightarrow \mathbb{R}_{\leq 1}[x], \quad T(f) = f',$$

where f' is the derivative of f . The usual basis in $\mathbb{R}_{\leq 2}[x]$ is

$$\beta = \{1, x, x^2\}.$$

The coordinate map is

$$S : \mathbb{R}_{\leq 2}[x] \rightarrow \mathbb{R}^3, \quad S(a_0 + a_1x + a_2x^2) = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}.$$

Then

$$STS^{-1} = Ax,$$

where

$$\begin{aligned} A &= \begin{bmatrix} [T(1)]_\beta & [T(x)]_\beta & [T(x^2)]_\beta \end{bmatrix} \\ &= \begin{bmatrix} [0]_\beta & [1]_\beta & [2x]_\beta \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Question. What is the smallest k with $A^k = 0$?

$$k = 3.$$

Example. Let C be an $n \times n$ matrix. Define

$$T : M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R}), \quad T(B) = CB.$$

Is T linear?

$$\begin{aligned} T(c_1 B_1 + c_2 B_2) &= C(c_1 B_1 + c_2 B_2) \\ &= c_1 C B_1 + c_2 C B_2 \\ &= c_1 T(B_1) + c_2 T(B_2). \end{aligned}$$

How do we compute $\dim \ker T$ and $\dim \operatorname{Im} T$?

Subquestion exercise. Let $n = 2$ and

$$C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

A basis of $M_{2 \times 2}(\mathbb{R})$ is

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

For this basis,

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

$$\begin{aligned}
T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) &= \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}, \\
T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) &= \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix}, \\
T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) &= \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix}.
\end{aligned}$$

With the coordinate map $S : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4$,

$$Ax = STS^{-1}(x), \quad A = \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}.$$

Therefore,

$$\dim \ker T = \dim \text{Nul } A, \quad \dim \text{Im } T = \text{rank } A.$$

Subquestion. If C is invertible, is $T(B) = CB$ invertible?

Yes.

$$T^{-1}(B) = C^{-1}B, \quad \ker T = \{0\}.$$

Example. Suppose V has basis

$$\beta = \{v_1, \dots, v_n\}$$

and another basis

$$\gamma = \{u_1, \dots, u_n\}.$$

The coordinate-change map is represented by

$$RS^{-1}(x) = Ax, \quad Ae_i = RS^{-1}(e_i).$$

The change of coordinates is represented by

$$\begin{array}{ccc} V & \xrightarrow{S} & \mathbb{R}^n \\ \text{id} \downarrow & & \downarrow RS^{-1} \\ V & \xrightarrow{R} & \mathbb{R}^n. \end{array}$$

Example. $\mathbb{R}^2$ has two bases

$$\beta = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

and

$$\gamma = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

The two coordinate maps are displayed as

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{S} & \mathbb{R}^2 \\ \text{id} \downarrow & & \downarrow RS^{-1} \\ \mathbb{R}^2 & \xrightarrow{R} & \mathbb{R}^2. \end{array}$$

Here

$$S^{-1}(e_1) = v_1, \quad S^{-1}(e_2) = v_2, \quad S^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix},$$

while

$$R^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}.$$

Therefore the coordinate-change matrix from β -coordinates to γ -coordinates is

$$RS^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix}.$$